

*A Report
on*

ONLINE MEET EXPRESSION ANALYZER

*carried out as part of the course Project Based Learning- 2
Submitted by*

***Deep Mundra
2427030793
IV Semester***

*in partial fulfilment for the award of the degree
of*

**BACHELOR OF TECHNOLOGY
In
Computer Science & Engineering**



**Department of Computer Science & Engineering,
School of Computer Science and Engineering,
Manipal University Jaipur,
*Jan-May 2026***

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Under the Guidance of :

Guide Name : Dr. Sakshi Shringhi

Guide Signature (with date) : _____

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CERTIFICATE

This is to certify that the project entitled "Online Meet Expression Analyzer" is a bonafide work carried out as PBL-2 End Term Assessment in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, by Deep Mundra bearing registration number 2427030793, during the academic semester IV of year 2025-2026.

Place: Manipal University Jaipur, Jaipur

Name of the project guide: _____

Signature of the project guide: _____

Abstract

The Online Meet Expression Analyzer is a real-time facial expression recognition system designed for virtual meeting environments. The system captures live video feeds, detects participant faces, extracts facial features through deep neural networks, and classifies emotional states into seven discrete categories: happiness, sadness, anger, surprise, fear, disgust, and neutral. The solution addresses three primary objectives: providing real-time individual emotion recognition with minimal latency (below 200ms), generating aggregate engagement metrics across multiple participants, and ensuring privacy-preserving implementation through on-device processing and consent-based deployment. The system employs a MobileNetV2-based CNN fine-tuned on the AffectNet dataset, integrated with a MTCNN detection pipeline and DeepSORT multi-person tracker, exposed through a React.js dashboard backed by a Python/Flask API. The project targets meeting hosts, educators, and tele-health professionals who lack real-time non-verbal feedback in virtual settings.

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1 Introduction

1.1 Objective of the Project

The primary objective of this project, as proposed in the synopsis, is to develop an intelligent, real-time facial expression analysis system specifically designed for online meeting environments. The system aims to:

- Provide real-time individual emotion recognition with end-to-end latency below 200 milliseconds.
- Generate aggregate engagement metrics across multiple participants (up to 10 simultaneous faces) over time.
- Ensure privacy-preserving implementation through complete on-device processing, explicit opt-in consent, and no persistent storage of facial imagery.
- Deliver an intuitive, web-based analytics dashboard providing meeting hosts with real-time emotion indicators, sentiment trends, and attention alerts.

1.2 Brief Description of the Project

Meeting hosts, educators, and presenters face considerable difficulty in assessing audience engagement, comprehension, and emotional responses during online sessions. Unlike physical classrooms or conference rooms where instructors can scan the room and immediately gauge student attention through facial expressions and body language, virtual environments present participants in grid layouts with small video windows, making comprehensive emotional assessment practically impossible for human observers.

This limitation manifests in several critical scenarios. In online education, instructors cannot determine whether students are confused, bored, or engaged, preventing adaptive teaching strategies. In business meetings, presenters lack feedback on whether their message resonates with stakeholders. In remote team collaboration, managers struggle to assess team morale and individual well-being. Mental health professionals conducting teletherapy face challenges in reading subtle emotional cues that inform diagnosis and treatment.

The Online Meet Expression Analyzer addresses these gaps by automatically analyzing video streams in real-time, detecting and classifying emotional states with accuracy approaching human-level performance in controlled conditions. The system captures live video feeds, detects participant faces via MTCNN, tracks identities using DeepSORT, extracts facial features through a MobileNetV2-based CNN, and classifies emotional states. Results are aggregated and presented to the meeting host through an intuitive React.js dashboard connected to a Python/Flask backend over WebSockets.

1.3 Technology Used

1.3.1 Hardware Requirement

Component	Minimum Specification
Processor	Intel Core i5 / AMD Ryzen 5 (quad-core, 2.0 GHz+)
RAM	8 GB (16 GB recommended)
Webcam	720p HD webcam (1080p recommended)
GPU	Optional — NVIDIA GPU with CUDA (speeds up inference)
Storage	5 GB free disk space
Network	Broadband internet for WebSocket signalling

1.3.2 Software Requirement

Category	Technology / Tool
Operating System	Windows 10 / macOS 11 / Ubuntu 20.04+
Programming Language	Python 3.8+, JavaScript (ES6+)
Deep Learning Framework	TensorFlow 2.x
Computer Vision	OpenCV 4.5+
Face Detection	MTCNN
Multi-Person Tracking	DeepSORT
Model Architecture	MobileNetV2 (ImageNet pre-trained, fine-tuned on AffectNet)
Backend Framework	Python Flask + Flask-SocketIO
Frontend Framework	React.js (no build step), Tailwind CSS, ShadCN UI
Real-time Communication	WebSockets
Database / Storage	Session-based in-memory only (no persistent facial data)
Development Tools	VS Code, Git, npm

2 Literature Review

A thorough survey of existing literature was conducted to identify state-of-the-art methods in facial expression recognition (FER), multi-person tracking, and application-oriented online meeting systems.

2.1 Evolution of Facial Expression Recognition

Corneanu et al. (IEEE TPAMI, 2016) provide a comprehensive survey tracing the evolution of FER methodologies from traditional computer vision approaches (HOG, LBP) to modern deep learning systems. The survey systematically categorizes methods and identifies key challenges, though it pre-dates widespread adoption of attention mechanisms and does not address real-time video conferencing specifics.

2.2 Lightweight Attention Models for FER

Khan et al. (IEEE Transactions on Affective Computing, 2023) propose E-VIT, a mobile-friendly Vision Transformer using an efficient attention mechanism to capture both local and global facial features. It achieves high accuracy with low computational cost and is designed for mobile/edge devices. However, real-time multi-face performance is not fully benchmarked.

2.3 Advanced FER Datasets

AffectReal (ACM Multimedia, 2024) is a novel dataset captured from online meeting scenarios including screen-sharing and varied lighting, with labels for engagement and confusion beyond basic emotions. While highly relevant, its access may be restricted and it is not yet widely adopted as a benchmark.

2.4 Benchmark Datasets

Mollahosseini et al. (IEEE Trans. Affective Computing, 2019) present AffectNet, providing over 1 million manually annotated images with both categorical emotion labels and continuous valence-arousal dimensions. The dataset improves diversity but requires substantial computational resources and has class imbalance and licensing restrictions.

2.5 Advanced Recognition Architectures

Zhao et al. (CVPR, 2016) proposed adaptive channel-wise feature learning for streaming video, optimized for real-time streaming. Sandler et al. (CVPR, 2018) introduced MobileNetV2 with inverted residual blocks for excellent efficiency-accuracy tradeoffs enabling real-time CPU inference. He et al. (CVPR, 2016) introduced ResNet with residual connections, achieving high accuracy at greater computational cost.

2.6 Application-Oriented Systems

Li et al. (HCI, 2020) developed a prototype integrating FER into a video conferencing platform, demonstrating practical feasibility and validating utility for hosts. Privacy concerns, however, were inadequately addressed.

2.7 Multi-Person Tracking

Wojke et al. (IEEE ICIP, 2017) present DeepSORT, extending SORT tracking with deep appearance features for improved identity maintenance, maintaining consistent IDs across occlusions with real-time performance. Parameter tuning is necessary and ID switches can occur in crowded scenarios.

2.8 Research Gaps Identified

Despite substantial progress, several challenges remain: (1) limited focus on meeting-specific constraints, (2) insufficient attention to privacy-preserving architectures, (3) lack of cultural sensitivity, (4) minimal investigation of temporal aggregation, and (5) inadequate ethical frameworks.

2.9 Comparative Study of Architectures

Parameter	Traditional	CNN (ResNet50)	Lightweight	CNN-LSTM	Proposed
Accuracy (FER2013)	60-70%	75-85%	70-80%	80-90%	75-85%
Inference Speed (FPS)	30+	10-15	25-30	5-8	20-25
Model Size (MB)	<1	98	14	120+	15-20
Hardware	CPU only	GPU	CPU	GPU	CPU/GPU
Real-time	Excellent	Moderate	Excellent	Poor	Excellent
Multi-person	Limited	Yes	Yes	Limited	Yes (DeepSORT)
Privacy	N/A	Poor	Poor	Poor	On-device
Power	Very Low	High	Low	Very High	Low-Moderate

MobileNetV2 provides the optimal balance: (1) 70-80% accuracy sufficient for practical assessment, (2) 25-30 FPS throughput, (3) 14 MB model size, (4) CPU-capable inference, and (5) strong initialisation from transfer learning.

3 Methodology and Framework

3.1 System Architecture

The system follows a modular pipeline with five stages: (1) Video Acquisition, (2) Face Detection and Tracking, (3) Feature Extraction, (4) Emotion Classification, and (5) Analytics and Visualisation. Parallel processing streams handle multiple faces concurrently. The React.js frontend communicates with the Flask backend over WebSockets for sub-200ms round-trip latency.

3.2 Algorithms and Techniques

Component	Implementation Details
Video Acquisition	OpenCV VideoCapture API for webcam input; frames extracted at 5-10 FPS.
Detection & Localisation	MTCNN (3-stage cascade of CNNs) for face detection and landmark localisation.
Multi-Person Tracking	DeepSORT combining Kalman filtering with deep cosine metric for identity maintenance.
Preprocessing & Alignment	Faces cropped, resized to 96x96, aligned via affine transform; photometric normalisation applied.
Feature Extraction	MobileNetV2 (ImageNet pre-trained) using depthwise separable convolutions for efficiency.
Emotion Classification	1280-D feature vector fed to FC layers with ReLU and softmax predicting 7 emotion categories.
Temporal Smoothing	Exponential Moving Average (EMA) filter reduces jitter between consecutive frames.
Visualisation (UI)	React.js frontend + Flask backend; analytics streamed via WebSockets.
Privacy & Ethics	All processing on-device; no facial data transmitted; explicit opt-in consent required.

3.3 Problem Statement

In online meeting environments, hosts and educators face significant challenges in assessing participant engagement, emotional states, and comprehension due to the absence of natural non-verbal communication cues. The screen-mediated nature of video conferencing prevents effective real-time emotional assessment at scale. Consequently, instructors cannot adapt teaching, presenters lack audience feedback, and managers struggle to evaluate group dynamics. An automated, real-time facial expression recognition system is required that can analyse participant emotions, provide actionable insights via intuitive dashboards, operate with minimal latency on consumer hardware, and ensure privacy-preserving implementation, thereby bridging the emotional communication gap.

4 Work Done

4.1 Input / Output Form Design

The system provides two primary user-facing interfaces:

Live Analysis Page: The webcam feed is displayed in real time with bounding boxes drawn around detected faces and the dominant emotion label overlaid. A sidebar panel shows session duration, frames processed, and current FPS. The user can adjust the detection rate (every 1s, 2s, or 5s), the confidence threshold (0-100%), and toggle the face detection overlay. A scrollable timeline records emotion entries with timestamps. Controls include Pause, Capture Frame, Toggle Overlay, and Stop.

Analytics Dashboard: Provides aggregate statistics including total sessions, frames analysed, average confidence, and average session duration. Charts include Emotion Trends Over Time (line/area), Average Confidence by Emotion (bar), and Emotion Distribution (pie). The dashboard supports week/month filtering and manual refresh.

4.2 Results and Discussion

The system was tested on a mid-range consumer laptop (Intel Core i5, 8 GB RAM, integrated graphics). The following performance metrics were recorded:

Metric	Target	Achieved
End-to-end Latency	< 200 ms	~120-180 ms (1 face)
Throughput (FPS)	20-25 FPS	20-25 FPS (detection every 1s)
Emotion Accuracy (7-class)	> 75%	~76% on FER2013 validation
Max Concurrent Faces	10	Up to 10 (tested with 4)
Setup Time	< 2 minutes	~90 seconds
Facial Data Transmission	Zero	Zero (verified via network monitor)

The dominant detected emotion during testing was consistently the correct label in well-lit, front-facing conditions. Accuracy dropped under low-light or heavy occlusion (face mask, glasses). The EMA temporal smoothing filter successfully reduced label jitter between consecutive frames.

5 Testing & Tools Used

Functional testing was carried out manually across the following test cases:

Test Case	Input Condition	Expected Output	Result
Single face, good lighting	1 participant, bright room	Correct emotion label, FPS ~20	Pass
Multiple faces (4)	4 participants in grid	Distinct bounding boxes, individual labels	Pass
Low-light condition	Dim room lighting	Face detected, reduced confidence	Partial Pass
Occluded face (glasses)	Participant wearing glasses	Face detected, neutral/sad bias noted	Partial Pass
No face in frame	Camera covered	No detection, no crash	Pass
Session stop and save	User clicks Stop	Session saved to dashboard history	Pass
Privacy check	Network traffic monitored	No facial data packets transmitted	Pass

Tools used include Python 3.10, TensorFlow 2.12, OpenCV 4.8, Flask 3.0, Flask-SocketIO, React 18, Tailwind CSS, ShadCN UI, and browser DevTools for network monitoring.

6 Conclusion and Future Work

The Online Meet Expression Analyzer successfully demonstrates real-time facial expression recognition in a virtual meeting context. The system achieves the targeted latency, accuracy, and privacy objectives on consumer hardware. The modular pipeline architecture allows independent component upgrades without redesigning the full system. Future work includes: (1) replacing MobileNetV2 with a fine-tuned Vision Transformer (E-ViT) for higher accuracy, (2) expanding the emotion taxonomy to include compound emotions and engagement scores, (3) integrating with native meeting platforms (Google Meet, Zoom) via browser extensions, (4) adding cultural sensitivity calibration for diverse participant populations, (5) implementing role-based access so individual emotion data remains private while only aggregate statistics are shared with the host.

7 Outcome

The project is currently at a functional prototype stage. A research paper documenting the system design, comparative architecture analysis, and performance benchmarks is being drafted for submission to an IEEE conference on Human-Computer Interaction or Affective Computing. The frontend dashboard and backend API are version-controlled on GitHub. Deployment to a cloud environment (Docker + Flask on a GPU-enabled server) is planned as a future milestone once the model accuracy is improved to exceed 80%.

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